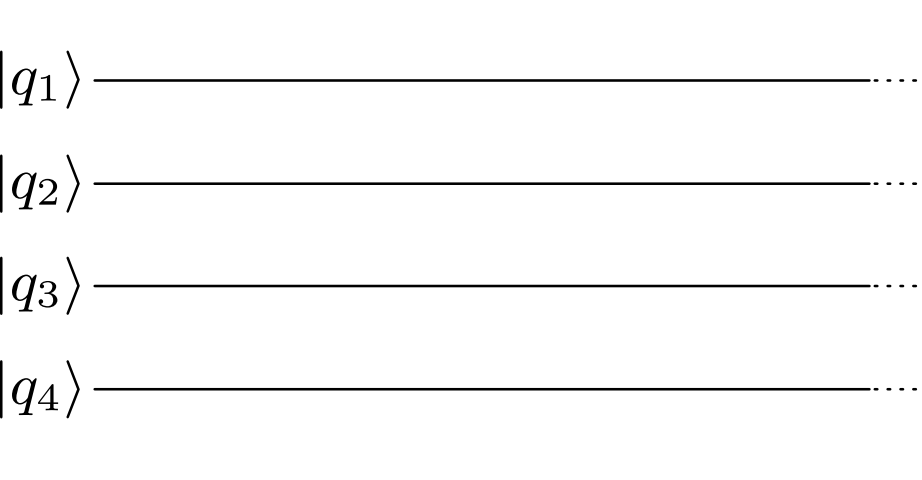
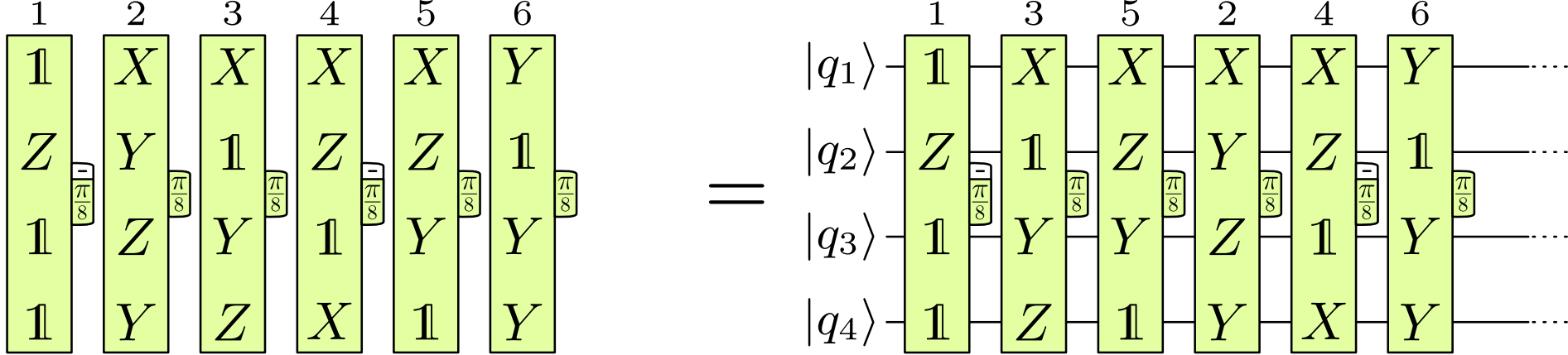

$|q_1\rangle$
$|q_2\rangle$
$|q_3\rangle$
$|q_4\rangle$


1 2 3 4 5 6
𝟙 X X X X Y
Z Y 𝟙 Z Z 𝟙
𝟙 Z Y 𝟙 Y Y
𝟙 Y Z X 𝟙 Y
-π/8
π/8
=
1 3 5 2 4 6
|q1⟩
|q2⟩
|q3⟩
|q4⟩